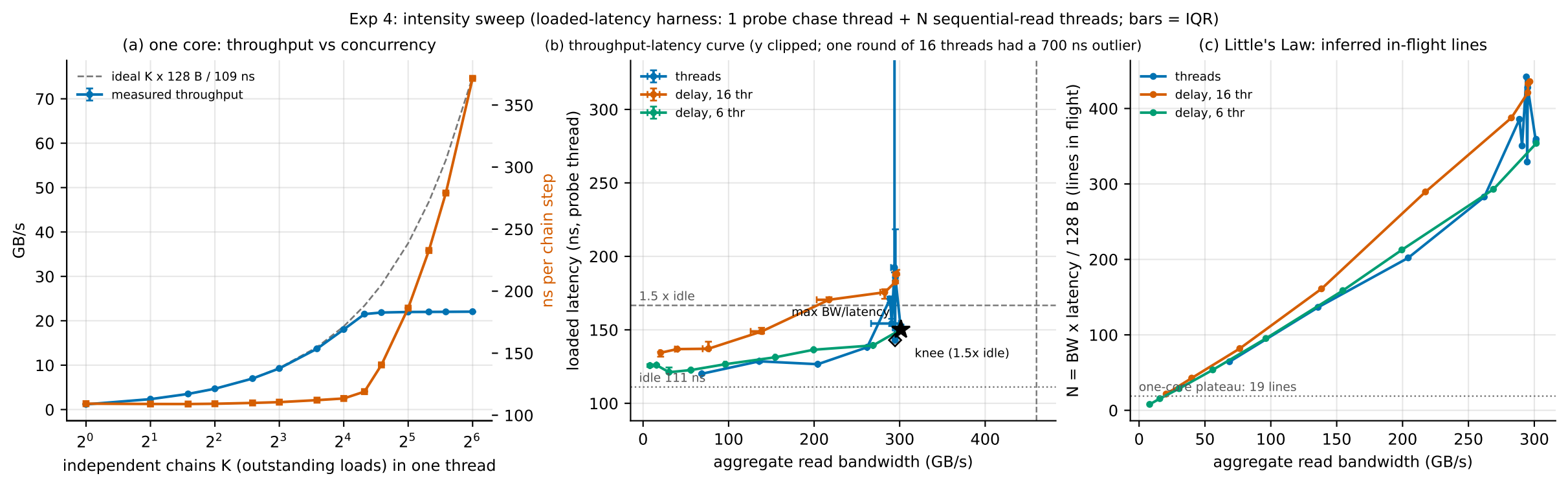
Exp 4: intensity sweep (loaded-latency harness: 1 probe chase thread + N sequential-read threads; bars = IQR)
(a) one core: throughput vs concurrency
ideal K x 128 B / 109 ns
measured throughput
GB/s
ns per chain step
independent chains K (outstanding loads) in one thread
(b) throughput-latency curve (y clipped; one round of 16 threads had a 700 ns outlier)
threads
delay, 16 thr
delay, 6 thr
loaded latency (ns, probe thread)
1.5 x idle
max BW/latency
knee (1.5x idle)
idle 111 ns
aggregate read bandwidth (GB/s)
(c) Little's Law: inferred in-flight lines
threads
delay, 16 thr
delay, 6 thr
N = BW x latency / 128 B (lines in flight)
one-core plateau: 19 lines
aggregate read bandwidth (GB/s)